Computer Organization II

Benchmarking Project

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COMPUTER SPECIFICATIONS

1. The brand of CPU (Intel or AMD): **Apple Silicon**

2. The model of CPU: **Apple M3**

3. The number of cores on CPU: **8-core CPU (4 performance cores +**

4 efficiency cores)

4. The clock rate of the CPU in GHz: **up to ~3.2 GHz**

5. The amount of memory in GB: **8 GB**

6. The speed of memory: **100 GB/s memory bandwidth**

7. The capacity of hard drive: **245.11 GB**

8. The type of hard drive: **SSD**

    a. Max sequential read speed: **2872.41 MB/s**
    b. Max sequential write speed: **1960.67 MB/s**
    c. Max random read speed: **59.18 MB/s**
    d. Max random write speed: **20.01 MB/s**

Benchmark Results

Benchmark	**Benchmark Recorded Time (s)**
32-bit Integer operation benchmark	7.4203 seconds
64-bit Floating point	16.7316 seconds
Memory benchmark	2.6856 seconds
Hard drive benchmark 1	0.960803 seconds
Hard drive benchmark 2	0.556430 seconds

FINAL BENCHMARK

The calculation of a single number using geometric means to summarize the results of all five benchmarks.

Geometric Mean =

$$\sqrt[5]{Benchmark\ 1\ \times\ Benchmark\ 2\ \times\ Benchmark\ 3\ \times\ Benchmark\ 4\ \times\ Benchmark\ 5}$$

$$= \sqrt[5]{7.4203\ \times\ 16.7316\ \times\ 2.6856\ \times\ 0.960803\ \times\ 0.556430}$$

= **2.81974**

SCREENSHOTS FOR BENCHMARK TIME

1. 32-bit Integer operation benchmark

```
avvhix@Avions-MacBook-Air benchmark % ./integer_benchmark
Benchmark 1:
32-bit Integer Operation Benchmark
This includes additions, multiplications, and divisions.

Execution Time: 7.4203 seconds
```

2. 64-bit Floating point operation benchmark

```
avvhix@Avions-MacBook-Air benchmark % ./float_benchmark
Benchmark 2:
64-bit Floating point operation benchmark
Which includes additions, multiplication, and division.
Execution Time: 16.7316 seconds
```

3. Memory benchmark

```
avvhix@Avions-MacBook-Air benchmark % ./memory_benchmark
Benchmark 3:
Memory benchmark
Which includes reading from, and writing to array, 4 bytes each time
Execution Time: 2.6856 seconds
```

4. Hard drive benchmark 1

```
avvhix@Avions-MacBook-Air benchmark % ./hard_drive_benchmark_1
Benchmark 4:
Hard Drive benchmark 1
Which includes reading from, and writing to file, 100 bytes each time
Execution Time: 0.960803 seconds
```

5. Hard drive benchmark 2

```
avvhix@Avions-MacBook-Air benchmark % ./hard_drive_benchmark_2
Benchmark 5:
Hard Drive benchmark 2
Which includes reading from, and writing to file, 10000 bytes each time
Execution Time: 0.556430 seconds
```